Chunxiao (Calvin) Lin

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Education

Virginia Tech, Ph.D. in Computer Engineering GPA: 4.0/4.0 Sep 2021 – May 2027 (Expected)

- Research Field: Neuromorphic computing, FPGA design, Wireless communication, Machine learning
- Focus: FPGA acceleration of ML models (**Echo State Network (ESN)**) for Next-G wireless communication

University of Southern California, Master of Engineering GPA: 3.95/4.0 Sep 2019 – May 2021

- Courses: Computer Architecture, VLSI Design, Digital System Design

Technologies

Languages: Verilog, SystemVerilog, Python, MATLAB, C

Tools: Xilinx Vivado, Vitis, Quartus, PyCharm, Jupyter Notebook, MATLAB, CLion, Cadence, NS3

HW Platforms: Xilinx Virtex-7 FPGA, Xilinx Zynq-7000 SoC FPGA

Projects

Real-Time 10G Ethernet PC-FPGA Testbed for Distributed MIMO Systems Feb 2026 – Jul 2026

- Designed and implemented an RTL-based two-mode windowed echo state network (WESN) accelerator on the ZCU216 RFSoC platform, achieving ultra-low-latency MIMO channel prediction.
- Developed an end-to-end real-time channel prediction testbed integrating the FPGA accelerator with a host PC through a 10G Ethernet interface for high-throughput data transmission.

Hardware-software Co-design of an On-chip ESN Apr 2025 – Sep 2025

- Designed and implemented an on-chip ESN with PL-based inference and PS-based learning on Xilinx Zynq-7000 SoC FPGA.

FPGA-based ESN with On-chip RLS Learning for MIMO Channel Prediction Apr 2024 – Dec 2024

- Designed and implemented an on-chip ESN with systolic-array-based Output layer architecture and hardware-optimized recursive least square learning block.
- Both inference and training circuits are optimized for high-dimensional data processing in MIMO channel prediction task.

Hardware-efficient ESN Detector with Online Learning for Massive MIMO Aug 2023 – Dec 2023

- Introduced the ESN-based symbol detector for massive MIMO symbol detection.

FPGA-based ML Model Design for MIMO-OFDM Symbol Detection Jun 2022 – Nov 2022

- Proposed a DSP-based neuron architecture for acceleration of an ESN-based symbol detector in a MIMO-OFDM system.
- Build a Software-defined Radio/FPGA joined platform and validate our design in a 2x2 MIMO system.

Optimized Reservoir Node Architecture for FPGA-based RC Systems Oct 2021 – Feb 2022

- Proposed a low-cost architecture for the reservoir neuron of the ESN.
- Designed a bit-serial matrix multiplier and direct spatial implementation of the weight matrix, along with a fixed-point design of the nonlinear function.

Low Power 10T-SRAM Design for In-Memory Computing with Leakage Current Reduction Oct 2020 – Dec 2020

- Designed a 64x64 10T-SRAM array for in-memory computing using two asymmetric sense amplifiers.
- Reduced leakage current with high-Vt, transistor sizing, transistor stacking, and negative Vgs techniques.

Publications

- **C. Lin**, U. Sumaya, M. Farhan, and Y. Yi, "FPGA-based Reservoir Computing with On-chip Learning for MIMO Channel Prediction," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 2026. **Accepted**.
- S. S. Chowdhury, M. R. Sarkar, J. Ling, **C. Lin**, and Y. Yi, "SPARC: Spiking Plasticity-Adaptive Reservoir Computing with On-Chip Reward-Modulated Triplet STDP for Spectrum Sensing," *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, 2026. *Submitted*.
- **C. Lin**, U. S. Khan, L. Liu, and Y. Yi, "FPGA-based High-speed Reservoir Computing for MIMO Channel Prediction," in *Proc. 2026 International Conference on Computing, Networking and Communications (ICNC)*, Maui, HI, USA, 2026, pp. 465–470. doi: 10.1109/ICNC68183.2026.11417007.
- A. Liu, M. F. Azmine, **C. Lin**, and Y. Yi, "Reservoir Computing: Foundations, Advances, and Challenges Toward Neuromorphic

Intelligence," *AI*, vol. 7, no. 2, Art. no. 70, 2026. doi: 10.3390/ai7020070.

- R. Gaurav, S. Sinha, **C. Lin**, T. C. Stewart, L. Liu, and Y. Yi, "Multivariate Legendre-SNN on Loihi-2 for Time Series Classification and 5G Jamming Detection," *IEEE Transactions on Circuits and Systems for Artificial Intelligence*, 2026. doi: 10.1109/TCASAI.2026.3658587.
- A. Liu, **C. Lin**, D. Rosen, and Y. Yi, "Echo State Networks in Reservoir Computing: Foundations, Benchmarks, and Applications to Next-G Wireless Communication," in *Proc. 2025 IEEE International Conference on Data Mining Workshops (ICDMW)*, Washington, DC, USA, 2025, pp. 3076–3080. doi: 10.1109/ICDMW69685.2025.00405.
- K. Yang, Y. Hu, D. Chen, **C. Lin**, Y. Yi, P. A. Beerel, and P. Nuzzo, "Machine Learning-Enhanced Analysis and Design for Trustworthy Integrated Circuits," in *AI-Enabled Electronic Circuit and System Design: From Ideation to Utilization*. Cham, Switzerland: Springer, 2025, pp. 467–496. doi: 10.1007/978-3-031-71436-8_13.
- **C. Lin** and Y. Yi, "Enhancing FPGA CAD Flow with AI-Powered Solutions," in *AI-Enabled Electronic Circuit and System Design*, A. Iranmanesh and H. Sayadi, Eds. Cham, Switzerland: Springer, 2025. doi: 10.1007/978-3-031-71436-8_7.
- **C. Lin** et al., "Leveraging Neuro-Inspired AI Accelerator for High-Speed Computing in 6G Networks," *Frontiers in Computational Neuroscience*, vol. 18, Art. no. 1345644, 2023.
- **C. Lin**, M. F. Azmine, and Y. Yi, "Accelerating Next-G Wireless Communications with FPGA-Based AI Accelerators," in *Proc. 2023 IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, 2023.
- **C. Lin**, Y. Liang, and Y. Yi, "FPGA-based Reservoir Computing with Optimized Reservoir Node Architecture," in *Proc. 23rd International Symposium on Quality Electronic Design (ISQED)*, Santa Clara, CA, USA, 2022, pp. 1–6. doi: 10.1109/ISQED54688.2022.9806247.
- D. Chen, **C. Lin**, and P. A. Beerel, "GF-Flush: A GF(2) Algebraic Attack on Dynamically Secured Scan Chains," in *Proc. 2021 IEEE International Symposium on Defect and Fault Tolerance in VLSI and Nanotechnology Systems (DFT)*, 2021, pp. 1–6. doi: 10.1109/DFT52944.2021.9568356.